

Polarized SI-Membrane Cosmology, Opus II: Universal Orientation Sorting and a Geometric Yield Ansatz

Volodimir Slobodyanyuk
Independent researcher
vslobody@hotmail.com

June 24, 2026
DOI: [10.5281/zenodo.20991835](https://doi.org/10.5281/zenodo.20991835)

Abstract

Opus I formulated polarized membrane cosmology as a three-sector yield problem: a higher-dimensional pre-projection state is mapped into a visible branch A , a conjugate branch B , and a neutral relic sector. The present note sharpens the mechanism by naming the membrane S_l and by separating two ideas that should not be conflated. The Standard Model particle spectrum is treated as input; the new structure proposed here is a universal orientation label sorted by the S_l -membrane. Above a membrane crossing threshold, particles traverse the membrane freely and ordinary annihilation proceeds. Below the threshold, after most annihilation has already occurred, residual oriented excitations are projected into dynamically isolated branches, while a neutral topological relic channel $\mathcal{R}_{S_l}$ may remain. The dark-to-baryonic matter ratio, $\Omega_{\text{DM}}/\Omega_b \simeq 5.3$, is then the first quantitative target of the program. This note does not derive 5.3 from first principles; it defines the minimal orientation-sorting, threshold, and relic-yield structure such a derivation must supply.

1 Purpose and status of Opus II

Opus I proposed the schematic three-sector projection

$$\mathcal{H}_{\text{deep}} \longrightarrow \mathcal{H}_A \oplus \mathcal{H}_B \oplus \mathcal{H}_{\mathcal{R}}. \quad (1)$$

The present note revises that language in two ways. First, the membrane responsible for the projection will be denoted by S_l and its projection operator by Π_{S_l} . Second, the branch label is not treated as an arbitrary sign analogous to an inserted CP phase. It is tied to worldline orientation relative to the membrane embedding:

$$|+\rangle \equiv |\rightarrow t\rangle, \quad |-\rangle \equiv |\leftarrow t\rangle. \quad (2)$$

Thus the relevant projection is not a nameless binary split, but a sorting of forward- and backward-time oriented excitations. References to e, μ, τ are illustrative only: the Standard Model already exhibits three generations across charged leptons, neutrinos, up-type quarks, and down-type quarks. In this note those family labels are not derived and need not be unified. The S_l membrane acts on the orientation label common to all these excitations, while preserving their Standard Model gauge identities.

This note remains an ansatz paper, not a completed calculation. A completed calculation would specify the S_l geometry, its threshold scale, its spectrum, and its topological indices. The specific quantitative target is

$$\frac{\Omega_{\text{DM}}}{\Omega_b} \approx 5.3, \quad (3)$$

with the additional requirement that the observed baryon-to-photon ratio remain

$$\eta_A = \frac{n_{b,A}}{n_{\gamma,A}} \sim 6 \times 10^{-10}. \quad (4)$$

The purpose of Opus II is to define the mathematical object that must eventually produce both numbers without independent tuning.

2 The S_l membrane and threshold projection

The membrane is assumed to be present throughout the relevant early-universe epoch. It is not switched on from nothing. Instead, its effective opacity depends on the energy of excitations crossing it. At high energy,

$$E \gg E_{S_l}, \quad (5)$$

particles and antiparticles traverse the S_l membrane in both directions with high transmission probability:

$$\mathcal{T}_{S_l}(E) \simeq 1. \quad (6)$$

During this transparent regime, ordinary annihilation proceeds and produces the photon bath. The membrane does not prevent the early annihilation era.

As the universe expands and cools, the characteristic kinetic energy of the residual excitations falls. When

$$E \lesssim E_{S_l}, \quad (7)$$

the membrane becomes effectively opaque to cross-orientation traversal. This threshold condition replaces the vague statement that the membrane “turns on.” More precisely,

$$\mathcal{T}_{S_l}(E) \rightarrow 0 \quad \text{for cross-sector motion as } E/E_{S_l} \rightarrow 0. \quad (8)$$

The projection temperature T_{S_l} is defined by the epoch at which the typical residual energy falls below the crossing threshold:

$$\langle E(T_{S_l}) \rangle \simeq E_{S_l}. \quad (9)$$

The membrane was present before T_{S_l} , but the particles could freely traverse it; after T_{S_l} , the remaining residual excitations are trapped into branch sectors according to their orientation.

The branch projection is then

$$\Pi_{S_l} | \rightarrow t \rangle = + | \rightarrow t \rangle, \quad (10)$$

$$\Pi_{S_l} | \leftarrow t \rangle = - | \leftarrow t \rangle. \quad (11)$$

Branch A is the sector receiving the forward-oriented residuals relative to our observed time orientation,

$$\mathcal{H}_A \sim P_A \mathcal{H}_{\text{deep}}, \quad P_A = \frac{1 + \Pi_{S_l}}{2}, \quad (12)$$

and branch B receives the conjugate orientation,

$$\mathcal{H}_B \sim P_B \mathcal{H}_{\text{deep}}, \quad P_B = \frac{1 - \Pi_{S_l}}{2}. \quad (13)$$

This does not require committing to the view that time is fundamentally thermodynamic. The label is kinematic/geometric: it is the orientation of a worldline crossing relative to the S_l membrane structure. Thermodynamic cooling only controls when the crossing threshold becomes dynamically relevant.

3 Relation to particle-antiparticle language

In the Feynman-Stueckelberg interpretation, antiparticles can be represented diagrammatically as particles with opposite time orientation. The present model promotes that orientation label into the branch-sorting variable of the S_l membrane. Schematically,

$$|\text{matter orientation}\rangle \sim |\rightarrow t\rangle, \quad (14)$$

while

$$|\text{antimatter orientation}\rangle \sim |\leftarrow t\rangle. \quad (15)$$

The statement is not that local positrons are forbidden in branch A . Local QED remains intact:

$$\gamma\gamma \leftrightarrow e^-e^+, \quad (16)$$

and ordinary annihilation is allowed. The projection concerns the large-scale residual population after the high-transmission annihilation epoch. It is a cosmological branch-occupation rule, not a local charge-superselection rule.

The visible branch must remain neutral under Standard Model gauge charges,

$$Q_A \simeq 0, \quad SU(3)_c \text{ singlet at macroscopic scales}, \quad (17)$$

and the relic channel must be Standard Model gauge neutral,

$$Q_{\text{SM}}(\mathcal{R}_{S_l}) = 0. \quad (18)$$

The S_l membrane sorts orientation and topological sector labels. It may not isolate unscreened electric or color charge.

4 Observed Standard Model generations and universal orientation sorting

The charged leptons

$$(e, \mu, \tau) \quad (19)$$

are used in this series only as the most transparent example of a broader empirical fact: Standard Model matter appears in three generations. The same threefold pattern is observed in

$$(e, \mu, \tau), \quad (\nu_e, \nu_\mu, \nu_\tau), \quad (u, c, t), \quad (d, s, b), \quad (20)$$

and, equivalently, in the three Standard Model families

$$\mathcal{G}_1 = (u, d, e, \nu_e), \quad (21)$$

$$\mathcal{G}_2 = (c, s, \mu, \nu_\mu), \quad (22)$$

$$\mathcal{G}_3 = (t, b, \tau, \nu_\tau). \quad (23)$$

This note does not need to explain why there are three generations. It uses the observed generation label as part of the state bookkeeping. The proposed S_l operation relevant to branch-relative matter asymmetry is not a free mixing of different Standard Model gauge representations, and it does not require a grand-unified embedding.

The minimal kinematic bookkeeping is

$$\mathcal{H}_{\text{observed}} = \bigoplus_f \left(\mathcal{H}_{\text{gen}}^{(f)} \otimes \mathcal{H}_{\text{orient}} \right), \quad (24)$$

where f labels an already-existing Standard Model representation or species class, and

$$\dim \mathcal{H}_{\text{gen}}^{(f)} = 3. \quad (25)$$

The family index $i = 1, 2, 3$ labels the observed generation within a fixed representation f . The orientation index $\sigma = \pm$ labels forward/backward worldline orientation. An observed excitation is therefore written schematically as

$$|f_i, \sigma\rangle, \quad i = 1, 2, 3, \quad \sigma = \pm. \quad (26)$$

The central S_l branch-sorting postulate is that the membrane acts on the orientation coordinate, not on the gauge-representation label:

$$\Pi_{S_l} |f_i, +\rangle = + |f_i, +\rangle, \quad (27)$$

$$\Pi_{S_l} |f_i, -\rangle = - |f_i, -\rangle. \quad (28)$$

Consequently,

$$P_A |f_i, +\rangle \simeq |f_i, +\rangle, \quad P_A |f_i, -\rangle \simeq 0, \quad (29)$$

$$P_B |f_i, -\rangle \simeq |f_i, -\rangle, \quad P_B |f_i, +\rangle \simeq 0, \quad (30)$$

for every observed Standard Model representation f and every generation label i . The same rule applies to charged leptons, neutrinos, up-type quarks, and down-type quarks. No step in this construction requires e to mix with u , μ to mix with c , or any other quark-lepton unification. Those particles keep their usual Standard Model gauge assignments. The membrane sorts their common orientation label.

The compact form is

$$|\Psi_f\rangle = \sum_{i=1}^3 \sum_{\sigma=\pm} c_{i\sigma}^{(f)} |f_i, \sigma\rangle, \quad (31)$$

with projectors

$$P_{i,\sigma,A}^{(f)} = P_i^{(f)} P_\sigma P_A. \quad (32)$$

Here $P_i^{(f)}$ selects the already-observed generation mode of representation f , P_σ selects worldline orientation, and P_A selects the visible branch. The point is not that S_l must derive the family

spectrum before the branch-asymmetry proposal can be stated. The point is that the observed family index and the proposed orientation index are independent labels carried by Standard Model excitations.

A stronger future version may attempt to derive the threefold family structure from an S_l spectral problem,

$$D_{S_l}\psi_n = \lambda_n\psi_n, \quad (33)$$

but that is not required for the present branch-sorting mechanism. The required object for Opus II is narrower: a membrane threshold and orientation projector that act universally on $\mathcal{H}_{\text{orient}}$ while preserving all Standard Model gauge charges.

5 Three-sector yield, restated

At the threshold epoch T_{S_l} , the post-annihilation residual state is projected as

$$\mathcal{H}_{\text{deep}} \xrightarrow{\Pi_{S_l, T_{S_l}}} \mathcal{H}_A \oplus \mathcal{H}_B \oplus \mathcal{H}_{\mathcal{R}}. \quad (34)$$

The branches carry opposite sector-relative baryon orientation,

$$B_A = -B_B, \quad (35)$$

while the global bookkeeping may remain balanced,

$$B_A + B_B = 0. \quad (36)$$

This is not ordinary global baryogenesis. It is branch-relative baryon asymmetry. Observers in A see a matter-dominated residual population; observers in B would describe their conjugate residual population using their own matter convention.

The stress-energy budget seen from branch A is schematically

$$\rho_{\text{obs},A} = \rho_{b,A} + \rho_{\gamma,A} + \rho_{\mathcal{R}} + \rho_{\Lambda} + \cdots, \quad (37)$$

where $\rho_{\mathcal{R}}$ is the energy density of compact, neutral topological relics produced by incomplete or residual S_l crystallization. The proposed target is

$$\frac{\rho_{\mathcal{R}}}{\rho_{b,A}} \approx \frac{\Omega_{\text{DM}}}{\Omega_b} \approx 5.3. \quad (38)$$

6 Why an arbitrary scalar potential is insufficient

A generic phenomenological order parameter potential,

$$V(\phi, T) = \frac{\lambda}{4} (\phi^2 - v^2(T))^2, \quad (39)$$

can describe a phase transition but cannot by itself explain the number 5.3. The parameters λ , $v(T)$, defect formation efficiency, and relic survival fraction may be tuned to reproduce many values. Such a model would fit the ratio, not derive it.

Therefore Opus II uses a threshold/index view. The desired result is not

$$\frac{\rho_{\mathcal{R}}}{\rho_b} = \text{chosen by parameters}, \quad (40)$$

but rather

$$\frac{\rho_{\mathcal{R}}}{\rho_b} = F_{S_l}[\mathcal{M}, \mathcal{B}, \mathcal{E}, \Gamma, E_{S_l}], \quad (41)$$

where $\mathcal{M}$ is the higher-dimensional membrane geometry, $\mathcal{B}$ is the branch structure, $\mathcal{E}$ denotes the internal excitation spectrum, Γ summarizes permitted topological transitions, and E_{S_l} is the S_l crossing threshold. A successful version requires this functional to be mostly fixed by topological or spectral data, not by adjustable couplings.

7 Geometric yield ansatz

Let I_b denote the topological multiplicity of branch-visible baryonic residues and I_R the topological multiplicity of neutral relic excitations. We write the conservative yield relation as

$$\frac{\rho_{\mathcal{R}}}{\rho_{b,A}} = \frac{m_R}{m_b} \frac{I_R}{I_b} \frac{\epsilon_R}{\epsilon_b}, \quad (42)$$

where:

- m_b is the characteristic baryonic mass per surviving branch residue;
- m_R is the characteristic mass of a neutral relic excitation;
- I_b and I_R are geometric or index-theoretic multiplicities;
- ϵ_b and ϵ_R are survival/transfer efficiencies after the S_l threshold transition.

Equation (42) shows where the theory must avoid cheating. If m_R , I_R/I_b , or ϵ_R/ϵ_b are arbitrary, then the ratio is fitted. If these quantities are fixed by the same S_l membrane geometry that defines the generation and orientation projectors, the ratio becomes predictive.

A geometry-dominated limit is obtained when the characteristic mass and survival factors are not free but determined by the same scale:

$$\frac{m_R}{m_b} \frac{\epsilon_R}{\epsilon_b} \rightarrow C_{S_l}, \quad (43)$$

with C_{S_l} fixed or slowly varying. Then

$$\frac{\rho_{\mathcal{R}}}{\rho_{b,A}} \approx C_{S_l} \frac{I_R}{I_b}. \quad (44)$$

A suggestive target is

$$\frac{16}{3} = 5.333\dots, \quad (45)$$

which is close to the observed dark-to-baryonic ratio. This paper does not claim that $16/3$ is derived. It is used only as a concrete target showing the type of topological quotient one would need.

8 A possible index structure

The intersecting-brane analogy suggests that visible matter multiplicities can arise from intersection numbers of wrapped cycles. Let Σ_a and Σ_b be cycles in the compact S_l geometry. The net localized branch-state multiplicity is modeled by an oriented intersection number,

$$I_b = \Sigma_a \circ \Sigma_b. \quad (46)$$

A neutral relic can instead correspond to closed defect loops, localized knots, or compact solitons of the branch order parameter. Its multiplicity may be associated with a different index,

$$I_R = \text{Ind}(D_{S_l})_{\text{neutral}}, \quad (47)$$

where D_{S_l} is an effective Dirac or stability operator coupled to the S_l membrane polarization background.

The speculative but testable idea is that both indices are derived from one spectral problem:

$$D_{S_l}\psi_n = \lambda_n\psi_n, \quad (48)$$

with different sectors corresponding to different eigenvalue classes:

$$\lambda_n \in \Lambda_b \rightarrow \text{branch baryonic residues}, \quad (49)$$

$$\lambda_n \in \Lambda_R \rightarrow \text{neutral topological relics}, \quad (50)$$

$$\lambda_n \in \Lambda_\sigma \rightarrow \text{forward/backward orientation labels}. \quad (51)$$

In the simplest finite-spectrum toy model,

$$\frac{I_R}{I_b} = \frac{\#(\Lambda_R)}{\#(\Lambda_b)}. \quad (52)$$

This is the mathematical missing object of the series. A later paper would need to propose a specific D_{S_l} and geometry for which the quotient becomes calculable.

9 The 5.3 target as a falsifiability condition

The ratio $\Omega_{\text{DM}}/\Omega_b$ gives the model a sharp target. The program fails if it can only say

$$\frac{\rho_{\mathcal{R}}}{\rho_b} = \text{whatever the parameters are set to}. \quad (53)$$

The program becomes nontrivial if a fixed S_l geometry yields

$$5 \lesssim F_{S_l} \lesssim 6 \quad (54)$$

without tuning. It becomes compelling only if it also yields the baryon-to-photon scale

$$\eta_A \sim 6 \times 10^{-10} \quad (55)$$

from the same crossing threshold condition

$$\langle E(T_{S_l}) \rangle \simeq E_{S_l}. \quad (56)$$

Thus the two quantitative targets are

$$\eta_A \sim 6 \times 10^{-10}, \quad (57)$$

$$\frac{\rho_{\mathcal{R}}}{\rho_{b,A}} \sim 5.3. \quad (58)$$

A useful model must not fit these independently. It must connect them through one S_l threshold scale or one topological functional.

10 Relic-sector constraints

The relic channel $\mathcal{R}_{S_l}$ is viable only if it satisfies

$$Q_{\text{SM}}(\mathcal{R}_{S_l}) = 0, \quad (59)$$

$$\tau_R \gg t_0, \quad (60)$$

$$\rho_R/\rho_b \approx 5.3, \quad (61)$$

$$G\mu_{\text{visible}} \lesssim 10^{-7} \quad \text{if visible string-like defects survive,} \quad (62)$$

$$\sigma_{R-\text{SM}} \ll \sigma_{\text{excluded}}. \quad (63)$$

The bound on $G\mu$ should be read as a warning. A horizon-scale network of visible cosmic strings is strongly constrained by CMB anisotropy data. The preferred relics in this program are compact, neutral, hidden-sector objects rather than ordinary electromagnetic or color flux tubes.

The correct analogy is not an unscreened electric string stretched across the visible universe. It is a compact topological excitation of the branch-separating S_l order parameter. Such an object may gravitate like dark matter while remaining neutral under Standard Model gauge interactions.

11 What Opus II does not claim

This paper does not claim:

1. a derivation of $\Omega_{\text{DM}}/\Omega_b = 5.3$;
2. a concrete compactification or brane construction;
3. a replacement for local QED or QCD;
4. a violation of Standard Model gauge neutrality;
5. an observable antimatter domain inside our branch;
6. a commitment to time as fundamentally thermodynamic;
7. a completed theory of baryogenesis.

Its claim is narrower: if polarized S_l -membrane cosmology is to become predictive, the dark-to-baryonic matter ratio and the baryon-to-photon ratio must be consequences of the same crossing threshold and spectral quotient, not independent fitted parameters.

12 Roadmap to Opus III

A useful next paper must choose one of the following routes:

1. **Threshold route.** Specify E_{S_l} and compute the residual branch population at $\langle E(T_{S_l}) \rangle \simeq E_{S_l}$.
2. **Intersecting-cycle route.** Specify cycles $\Sigma_a, \Sigma_b, \dots$ and compute visible residues and neutral relics from their intersection form.
3. **Index-operator route.** Define D_{S_l} and compute the quotient of zero modes or stable modes assigned to baryonic and relic sectors.

4. **Orientation route.** Derive the universal orientation projectors P_σ while preserving the already-observed Standard Model generation and gauge labels. A later, stronger version may also attempt to derive the threefold family spectra, but this is not required for the branch-sorting claim.
5. **Compact soliton route.** Construct an effective hidden order parameter whose solitons are stable, neutral, cold, and have an abundance tied to T_{S_l} .
6. **No-go route.** Prove that no such quotient can produce 5.3 while satisfying gauge neutrality, stability, threshold timing, and CMB bounds.

The no-go route is included deliberately. A useful theory program must be able to fail.

13 Conclusion

Opus II revises polarized membrane cosmology around the S_l membrane. The branch label is not an arbitrary sign: it is identified with forward/backward worldline orientation relative to the membrane. The membrane is present throughout the relevant era but becomes dynamically sorting only when residual particle energies fall below the S_l crossing threshold. The observed three generations of Standard Model matter are carried as empirical generation labels; e, μ, τ are examples, not the whole claim. The required S_l mechanism for this paper is a universal orientation sorting acting across all Standard Model representations without mixing their gauge identities, together with a compact neutral relic channel.

The proposed three-sector transition is therefore

$$\mathcal{H}_{\text{deep}} \xrightarrow{S_l} \mathcal{H}_A \oplus \mathcal{H}_B \oplus \mathcal{H}_R. \quad (64)$$

It is only physically useful if the visible baryon residue and the neutral relic density are co-produced by the common S_l spectral structure. The observed ratio

$$\Omega_{\text{DM}}/\Omega_b \approx 5.3 \quad (65)$$

therefore becomes a test of whether the S_l membrane has geometric content. If 5.3 is fitted, the model is only notation. If it follows from an index quotient or threshold spectrum, the model becomes a candidate mechanism linking branch-relative matter asymmetry, Standard Model family projection, and dark relic abundance.

References

- [1] N. Aghanim et al. [Planck Collaboration], “Planck 2018 results. VI. Cosmological parameters,” *Astron. Astrophys.* **641**, A6 (2020), arXiv:1807.06209.
- [2] P. A. R. Ade et al. [Planck Collaboration], “Planck 2013 results. XXV. Searches for cosmic strings and other topological defects,” *Astron. Astrophys.* **571**, A25 (2014), arXiv:1303.5085.
- [3] J. Lizarraga, J. Urrestilla, D. Daverio, M. Hindmarsh and M. Kunz, “New CMB constraints for Abelian Higgs cosmic strings,” *JCAP* **10**, 042 (2016), arXiv:1609.03386.
- [4] A. M. Uranga, “Chiral four-dimensional string compactifications with intersecting D-branes,” *Class. Quant. Grav.* **20**, S373 (2003), arXiv:hep-th/0301032.

- [5] R. Blumenhagen, M. Cvetič, P. Langacker and G. Shiu, “Toward realistic intersecting D-brane models,” *Ann. Rev. Nucl. Part. Sci.* **55**, 71–139 (2005), arXiv:hep-th/0502005.
- [6] M. Hindmarsh, R. Kirk and S. M. West, “Dark Matter from Decaying Topological Defects,” *JCAP* **03**, 037 (2014), arXiv:1311.1637.
- [7] S. Afach et al., “Search for topological defect dark matter with a global network of optical magnetometers,” *Nature Physics* **17**, 1396–1401 (2021).
- [8] R. Dick, “Very heavy dark Skyrmions,” *European Physical Journal C* **79**, 369 (2019).